

Chapter 4: Knowledge Representation, Inference & Reasoning

8 hours | 14 marks — *HIGHEST PRIORITY (tied with Ch.7)*

"Logic is the cement of our civilization with which we ascend from chaos using reason as our guide."

— Gene Roddenberry

The Hook — What Does It Mean to "Know" Something?

Your brain stores billions of facts:

- "Kathmandu is in Nepal"
- "If it's raining, I should carry an umbrella"
- "All humans are mortal"

And you can *reason* with these facts:

- From "Socrates is human" and "All humans are mortal" → "Socrates is mortal"

AI needs to do the same thing. But a machine can't hold informal knowledge — it needs a **formal language** for representing facts and a **formal system** for drawing conclusions.

This chapter is about exactly that: **formal logic as a tool for AI reasoning.**

4.1 Formal Logic — The Foundation

Why Logic?

Logic gives us:

1. **Precision** — no ambiguity

2. **Soundness** — if premises are true, conclusions must be true

3. **Completeness** — all true things can be proven

These properties make logic a foundation for AI systems that need to reason reliably.

Core Concepts

4.1.1 LOGICAL CONNECTIVES

These are the building blocks of logical statements:

Symbol	Name	Meaning	Example
$\neg$	NOT (Negation)	Opposite	$\neg P$ = "not P"
$\wedge$	AND (Conjunction)	Both true	$P \wedge Q$ = "P and Q"
$\vee$	OR (Disjunction)	At least one true	$P \vee Q$ = "P or Q"
$\rightarrow$	IMPLIES (Implication)	If P then Q	$P \rightarrow Q$
$\leftrightarrow$	IFF (Biconditional)	P if and only if Q	$P \leftrightarrow Q$

4.1.2 TRUTH TABLES

Truth tables show the value of a formula for all possible input values.

For $P \rightarrow Q$ (implication):

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

 **Why is $F \rightarrow T$ true?** This is called "vacuous truth." If the premise is false, the implication doesn't fire at all. Think: "If it rains, I'll carry an umbrella" — if it doesn't rain, I haven't broken my promise regardless of what I do.

For $P \leftrightarrow Q$ (biconditional):

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

4.1.3 TAUTOLOGY AND CONTRADICTION

Tautology: A formula that is **always true** regardless of variable values.

Contradiction: A formula that is **always false**. **Contingency:** Neither — sometimes true, sometimes false.

Examples:

- $P \vee \neg P$ — always true (tautology) — "Either it rains or it doesn't"
- $P \wedge \neg P$ — always false (contradiction) — "It rains and doesn't rain simultaneously"
- $P \rightarrow Q$ — depends on P and Q (contingency)

How to check: Build the truth table. If last column is all T $\rightarrow$ tautology. All F $\rightarrow$ contradiction.

4.1.4 WELL-FORMED FORMULAS (WFF)

A WFF is a valid, grammatically correct logical expression.

Rules:

- Any single atom (P, Q, R...) is a WFF
- If A is a WFF, so is $\neg A$
- If A and B are WFFs, so are: $A \wedge B$, $A \vee B$, $A \rightarrow B$, $A \leftrightarrow B$
- Nothing else is a WFF

Valid WFFs: P, $\neg Q$, $P \wedge Q$, $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$

Invalid: $\wedge P$, $Q \wedge \wedge R$, $\rightarrow P$

4.1.5 SYNTAX VS. SEMANTICS

Aspect	Syntax	Semantics
Meaning	Grammar — <i>how</i> to write it	Meaning — <i>what</i> it says
Focus	Form of the formula	Truth value of the formula
Example	" $P \wedge Q$ " is syntactically valid	Under $P=T, Q=F$, its value is F

4.2 Propositional Logic

What Is It?

Propositional logic deals with **propositions** — statements that are either true or false.

Propositions: "It is raining" (P), "The road is wet" (Q)

Not propositions: "Is it raining?" (question), "Close the door!" (command)

Key Laws

These are fundamental identities you must know:

Law	Formula
De Morgan's (AND)	$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
De Morgan's (OR)	$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
Double Negation	$\neg\neg P \equiv P$
Contrapositive	$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$
Modus Ponens	$P, P \rightarrow Q \vdash Q$
Modus Tollens	$\neg Q, P \rightarrow Q \vdash \neg P$

De Morgan's Law intuition: "Not (A and B)" = "Not A or Not B"

Think: "It's not true that (it's both raining AND sunny)" = "Either it's not raining OR it's not sunny"

Limitations of Propositional Logic

Propositional logic cannot express:

- **Generalizations:** "All birds can fly" — would need one proposition per bird
- **Relationships:** "Socrates is taller than Plato"
- **Quantification:** "There exists an x such that..."

This is why we need **Predicate Logic**.

4.3 Predicate Logic (First-Order Predicate Logic — FOPL)

The Big Upgrade

FOPL adds two powerful things to propositional logic:

1. **Objects** and **predicates** about them
2. **Quantifiers** ($\forall$ and $\exists$)

4.3.1 Syntax of FOPL

Terms:

- Constants: specific objects (John, Kathmandu, 5)
- Variables: stand for any object (x, y, z)
- Functions: map objects to objects (father(John), add(2,3))

Predicates: Express properties or relationships

- Unary: Human(Socrates) — "Socrates is human"
- Binary: Loves(John, Mary) — "John loves Mary"
- n-ary: Between(x, y, z) — "x is between y and z"

Formulas: Built with predicates, connectives, and quantifiers

4.3.2 Quantification

UNIVERSAL QUANTIFIER: $\forall$ (FOR ALL)

$\forall x P(x)$ = "For every object x, P(x) is true"

Example: $\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

= "For all x, if x is human, then x is mortal"

EXISTENTIAL QUANTIFIER: $\exists$ (THERE EXISTS)

$\exists x P(x)$ = "There exists some object x such that P(x) is true"

Example: $\exists x [\text{Loves}(\text{John}, x)]$

= "There exists someone that John loves"

QUANTIFIER INTERACTION

Formula	Meaning
$\forall x \forall y P(x,y)$	Everyone relates P to everyone
$\exists x \exists y P(x,y)$	Someone relates P to someone
$\forall x \exists y P(x,y)$	Everyone has someone they relate P to
$\exists x \forall y P(x,y)$	Someone relates P to everyone

⚠ Order matters! $\forall x \exists y \neq \exists x \forall y$

$\forall x \exists y \text{ Loves}(x,y)$ = "Everyone loves someone" (each person may love a different person)

$\exists x \forall y \text{ Loves}(x,y)$ = "There is someone who loves everyone" (much stronger!)

DE MORGAN'S FOR QUANTIFIERS

Formula	Equivalent
$\neg \forall x P(x)$	$\exists x \neg P(x)$
$\neg \exists x P(x)$	$\forall x \neg P(x)$

"Not everyone is tall" = "Someone is not tall"

"No one is perfect" = "Everyone is not perfect"

4.3.3 Worked Example: Translating English $\rightarrow$ FOPL

Sentence: "Every student who studies hard passes the exam."

Let:

- $\text{Student}(x)$ = "x is a student"
- $\text{Studies}(x)$ = "x studies hard"
- $\text{Passes}(x)$ = "x passes the exam"

Translation:

$\forall x [\text{Student}(x) \wedge \text{Studies}(x) \rightarrow \text{Passes}(x)]$

Sentence: "Some professors are both researchers and teachers."

Let:

- $\text{Prof}(x)$ = "x is a professor"
- $\text{Researcher}(x)$ = "x is a researcher"
- $\text{Teacher}(x)$ = "x is a teacher"

Translation:

$\exists x [\text{Prof}(x) \wedge \text{Researcher}(x) \wedge \text{Teacher}(x)]$

4.3.4 Interpretation in FOPL

An **interpretation** assigns meaning to:

- A **domain** (set of objects)
- Meaning to constants (which object each constant refers to)
- Meaning to predicates (which tuples satisfy them)
- Meaning to functions (what they compute)

Example: Domain = {people in a classroom}

Constant "Alice" → refers to Alice

Predicate $\text{Smart}(x)$ → {Alice, Bob, Charlie}

Under this interpretation: $\text{Smart}(\text{Alice})$ is TRUE

A formula is **valid** if it's true under ALL interpretations.

A formula is **satisfiable** if it's true under SOME interpretation.

4.4 Horn Clauses

What Are They?

A **Horn clause** is a special form of logical statement:

$$p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow q$$

Where:

- $p_1 \dots p_n$ are **positive** literals (the conditions)
- q is a **single positive** conclusion

Or equivalently in clause form: $\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n \vee q$

(At most ONE positive literal)

Why Horn clauses matter:

- They are the basis of **Prolog** (the AI programming language)
- Resolution on Horn clauses is efficient — **linear time!**
- Most real knowledge bases can be expressed as Horn clauses

Example Horn clauses:

```
parent(tom, bob)      ← fact (no conditions)
parent(bob, ann)      ← fact
ancestor(X, Y) :- parent(X, Y)      ← rule
ancestor(X, Y) :- parent(X, Z), ancestor(Z, Y) ← recursive rule
```

4.5 Rules of Inference

These are the valid patterns of reasoning — formal "moves" you can make:

The Essential Set

1. MODUS PONENS (MP) — MOST IMPORTANT!

```
P
P → Q
```


$$\frac{}{Q}$$

"If P is true, and P implies Q , then Q is true."

Example:

- It is raining. (P)
- If it rains, the road is wet. ($P \rightarrow Q$)
- Therefore: The road is wet. (Q) ✓

2. MODUS TOLLENS (MT)

$$\frac{\neg Q \quad P \rightarrow Q}{\neg P}$$

"If Q is false, and P would imply Q , then P must be false."

Example:

- The road is NOT wet. ($\neg Q$)
- If it rains, the road is wet. ($P \rightarrow Q$)
- Therefore: It is NOT raining. ($\neg P$) ✓

3. AND-INTRODUCTION

$$\frac{P \quad Q}{P \wedge Q}$$

4. AND-ELIMINATION

$$\frac{P \wedge Q}{P \quad (\text{or } Q)}$$

5. HYPOTHETICAL SYLLOGISM (CHAIN RULE)

$$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ \hline P \rightarrow R \end{array}$$

"If A leads to B, and B leads to C, then A leads to C."

6. DISJUNCTIVE SYLLOGISM

$$\begin{array}{l} P \vee Q \\ \neg P \\ \hline Q \end{array}$$

"Either P or Q. P is false. Therefore Q."

7. RESOLUTION (GENERALIZATION OF MP — KEY FOR AI!)

$$\begin{array}{l} P \vee Q \\ \neg P \vee R \\ \hline Q \vee R \end{array}$$

This is the most powerful rule — it's the basis of automated theorem proving!

4.6 Unification

What Is It?

Unification is the process of making two logical expressions identical by substituting variables.

It's like pattern matching — "Can these two expressions be made to match?"

How It Works

Given two atoms, find a **substitution** (θ) that makes them identical.

Example:

Unify: $f(x, b)$ and $f(a, y)$

Substitution: $\{x \leftarrow a, y \leftarrow b\}$

Result: $f(a, b) = f(a, b)$  **Unified!**

Another example:

Unify: $P(x, f(x))$ and $P(a, f(a))$

Substitution: $\{x \leftarrow a\}$

Result: $P(a, f(a)) = P(a, f(a))$ 

Failure case:

Unify: $P(a, b)$ and $P(a, c)$ — b and c are different constants $\rightarrow$ **FAIL** 

The Occur Check

The substitution $\{x \leftarrow f(x)\}$ would create an infinite loop: $x = f(x) = f(f(x)) = f(f(f(x))) \dots$

Never substitute a variable with a term that contains itself!

4.7 Resolution Refutation System (RRS)

The Big Idea

"To prove a statement is true, assume it's false and derive a contradiction."

This is called **Proof by Contradiction** or **Refutation**.

The Steps

1. **Negate** the goal (what you want to prove)
2. **Convert** all sentences to **Conjunctive Normal Form (CNF)**
3. **Apply Resolution** repeatedly until you get the empty clause (contradiction!)

4. **Conclusion:** The original goal is TRUE.

CNF Conversion Steps

To convert to CNF:

1. Eliminate $\leftrightarrow$ using: $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$
2. Eliminate $\rightarrow$ using: $P \rightarrow Q \equiv \neg P \vee Q$
3. Move $\neg$ inward using De Morgan's and double negation
4. Distribute $\vee$ over $\wedge$

Worked Example: RRS

Knowledge Base:

1. $\text{Human}(\text{Socrates})$ — "Socrates is human"
2. $\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$ — "All humans are mortal"

Goal to prove: $\text{Mortal}(\text{Socrates})$

Steps:

Step 1: Negate goal $\rightarrow \neg \text{Mortal}(\text{Socrates})$

Step 2: Convert KB to clause form:

- Clause 1: $\text{Human}(\text{Socrates})$
- Clause 2: $\neg \text{Human}(x) \vee \text{Mortal}(x)$ [from rule 2 after conversion]
- Clause 3: $\neg \text{Mortal}(\text{Socrates})$ [negated goal]

Step 3: Apply Resolution:

```
Resolve Clause 1 and Clause 2 (with x = Socrates):
Human(Socrates) +  $\neg \text{Human}(\text{Socrates}) \vee \text{Mortal}(\text{Socrates})$ 
 $\rightarrow \text{Mortal}(\text{Socrates})$ 

Resolve this with Clause 3:
Mortal(Socrates) +  $\neg \text{Mortal}(\text{Socrates})$ 
 $\rightarrow \square$  (empty clause — CONTRADICTION!)
```

Step 4: Contradiction found $\rightarrow$ Original goal **Mortal(Socrates)** is TRUE 

4.8 Answer Extraction from RRS

The RRS can also be modified to **extract answers** to queries, not just verify them.

Example:

Query: "Who is mortal?" = $\exists x \text{ Mortal}(x)$?

Modified approach: Add "answer literal" to negated goal.

Negate and add answer literal:

$\neg \text{Mortal}(x) \vee \text{Answer}(x)$

When resolution produces $\text{Answer}(\text{Socrates})$ — that's your answer!

4.9 Rule-Based Deduction Systems

Two Directions of Reasoning

FORWARD CHAINING (DATA-DRIVEN)

"Start from what you know. Apply rules. See what new facts you can derive. Repeat."

Known Facts:	Rules:	Derived:
$\{A, B\}$	$A \wedge B \rightarrow C$	$\{A, B, C\}$
	$C \rightarrow D$	$\{A, B, C, D\}$
	$A \wedge D \rightarrow E$	$\{A, B, C, D, E\} \leftarrow \text{GOAL!}$

Good for: **generating all consequences** of a set of facts.

BACKWARD CHAINING (GOAL-DRIVEN)

"Start from the goal. Work backward to see what facts would prove it."

```

Goal: Prove E
  ↓
What proves E? → A ∧ D → E
  ↓
Sub-goals: Prove A, Prove D
  
```

```

↓
A is a fact ✓. What proves D?  $\rightarrow C \rightarrow D$ 
↓
Sub-goal: Prove  $C \rightarrow C$  is a fact ✓
DONE!

```

Good for: **answering specific queries** — only explores relevant rules.

Comparison

Feature	Forward Chaining	Backward Chaining
Direction	Facts $\rightarrow$ Goals	Goals $\rightarrow$ Facts
Driven by	Available data	Query/Goal
Good for	Deriving all consequences	Answering specific questions
Used in	Expert systems, databases	Prolog, query answering

4.10 Statistical Reasoning — Probability and Bayes' Theorem

Why Probability in AI?

The world is **uncertain**. Logic gives us certainty. But:

- "Will it rain tomorrow?" — we can't know for sure
- "Does this patient have cancer?" — symptoms give evidence, not certainty

We need a way to **reason under uncertainty** $\rightarrow$ **Probability Theory**.

4.10.1 Basic Probability

```

P(A) = probability that event A occurs
0 ≤ P(A) ≤ 1

P(Ω) = 1 (something must happen)
P(∅) = 0 (impossible event)

```

Joint Probability: $P(A \wedge B) = P(A) \times P(B|A)$

Where $P(B|A)$ = conditional probability of B given A.

4.10.2 Bayes' Theorem — The Crown Jewel of Statistical AI

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

In words:

$$P(\text{Hypothesis} | \text{Evidence}) = \frac{P(\text{Evidence} | \text{Hypothesis}) \times P(\text{Hypothesis})}{P(\text{Evidence})}$$

Or using intuitive names:

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

This is the fundamental formula for **updating beliefs given new evidence**.

EXAMPLE: MEDICAL DIAGNOSIS

A disease affects 1% of the population. A test is:

- 99% accurate (if sick, 99% chance positive)
- 1% false positive rate (if healthy, 1% chance positive)

Question: If you test positive, what's the probability you're actually sick?

Using Bayes:

- $P(\text{Sick}) = 0.01$ (prior — disease rate)
- $P(+|\text{Sick}) = 0.99$ (likelihood — true positive rate)
- $P(+) = P(+|\text{Sick}) \times P(\text{Sick}) + P(+|\text{Healthy}) \times P(\text{Healthy}) = 0.99 \times 0.01 + 0.01 \times 0.99 = 0.0099 + 0.0099 = 0.0198$

$$P(\text{Sick}|+) = \frac{0.99 \times 0.01}{0.0198} = 0.0099 / 0.0198 \approx 50\%$$

Shocking result! Even with a 99% accurate test and a positive result, you're only ~50% likely to be sick!

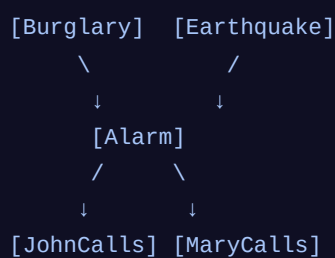
This is the **base rate fallacy** — the rarity of the disease matters enormously.

4.10.3 Causal Networks (Bayesian Networks)

A **Bayesian network** is a directed graph where:

- **Nodes** = random variables
- **Edges** = causal relationships ($A \rightarrow B$ means A causes B)
- Each node has a **conditional probability table (CPT)**

Example: Alarm Network (classic AI example)



Given probability tables:

- $P(\text{Burglary}) = 0.001$
- $P(\text{Earthquake}) = 0.002$
- $P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) = \text{various conditional probabilities}$

Query: $P(\text{Burglary} \mid \text{JohnCalls}, \text{MaryCalls})$?

This is what Bayesian networks excel at — **probabilistic inference** through a causal model.

Reasoning in Belief Networks

Three types of inference:

Type	Direction	Example
Causal	Parents $\rightarrow$ Children	Burglary $\rightarrow$ Alarm $\rightarrow$ JohnCalls

Type	Direction	Example
Diagnostic	Children $\rightarrow$ Parents	JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary
Intercausal	Between parents via child	Burglary $\leftrightarrow$ Earthquake (given Alarm)

"Explaining away" phenomenon: If we know the alarm is on AND we learn there was an earthquake, our belief in burglary goes DOWN — the earthquake "explains away" the alarm.



Math Intuition Corner

Why Resolution is Complete

The Resolution rule is **refutation complete** for first-order logic:

- If a set of clauses is unsatisfiable, resolution will derive the empty clause
- This takes at most exponential time in the worst case
- For Horn clauses: linear time!

The Connection Between Logic and Probability

Logic	Probability
P is True/False	$P(A) \in [0,1]$
$P \rightarrow Q$	$P(Q)$
$P \wedge Q$	$P(A \cap B)$
$P \vee Q$	$P(A \cup B)$

Probability is logic under uncertainty.



Video Resources

Topic	Video	Why Watch
Propositional Logic	Intro to Logic — Khan Academy	Interactive exercises
FOPL and Quantifiers	First Order Logic — Computerphile	Clear UK academic explanation
Bayes Theorem visual	Bayes theorem geometry — 3Blue1Brown	Absolutely stunning visual proof
Bayes theorem clear	Bayes Theorem — StatQuest	Clearest step-by-step ever
Bayesian Networks	Bayesian Networks explained	Practical and conceptual
Resolution in logic	Resolution principle explained	Worked examples

Worked Examples

Example 1: Truth Table Verification

Verify that the Contrapositive ($P \rightarrow Q \equiv (\neg Q \rightarrow \neg P)$) is a tautology:

P	Q	$P \rightarrow Q$	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Column 3 = Column 6 $\rightarrow$ **They are equivalent (tautology)** 

Example 2: FOPL Translation

English: "No student failed both Math and Science."

Step 1: Let:

- $\text{Student}(x)$, $\text{Failed_Math}(x)$, $\text{Failed_Science}(x)$

Step 2: "No student failed both" = "For all x, if x is a student, then not (x failed math AND x failed science)"

FOPL: $\forall x [\text{Student}(x) \rightarrow \neg(\text{Failed_Math}(x) \wedge \text{Failed_Science}(x))]$

Equivalent: $\forall x [\text{Student}(x) \rightarrow (\neg\text{Failed_Math}(x) \vee \neg\text{Failed_Science}(x))]$

Example 3: Resolution Step by Step

Prove: "John is mortal" given:

1. $\text{Human}(\text{John})$
2. $\forall x \text{Human}(x) \rightarrow \text{Mortal}(x)$

CNF conversion:

1. $\text{Human}(\text{John})$
2. $\neg\text{Human}(x) \vee \text{Mortal}(x)$
3. $\neg\text{Mortal}(\text{John})$ [negated goal]

Resolution:

- Step 1: Resolve (1) and (2) with $\{x \leftarrow \text{John}\}$: $\{\text{Human}(\text{John})\}$ and $\{\neg\text{Human}(\text{John}) \vee \text{Mortal}(\text{John})\} \rightarrow \{\text{Mortal}(\text{John})\}$
- Step 2: Resolve $\{\text{Mortal}(\text{John})\}$ and (3) $\{\neg\text{Mortal}(\text{John})\} \rightarrow \{\}$ empty clause! 

Conclusion: $\text{Mortal}(\text{John})$ is proved!

Exam-Ready Summary

Logical Connectives Quick Reference

Symbol	Name	P=T,Q=T	P=T,Q=F	P=F,Q=T	P=F,Q=F
$\neg P$	NOT	F	F	T	T
$P \wedge Q$	AND	T	F	F	F
$P \vee Q$	OR	T	T	T	F

Symbol	Name	P=T,Q=T	P=T,Q=F	P=F,Q=T	P=F,Q=F
$P \rightarrow Q$	IF-THEN	T	F	T	T
$P \leftrightarrow Q$	IFF	T	F	F	T

FOPL vs. Propositional Logic

Feature	Propositional	Predicate (FOPL)
Objects	No	Yes
Predicates	No	Yes
Quantifiers	No	$\forall$ and $\exists$
Expressiveness	Limited	Full (Turing complete)

Inference Rules to Memorize

1. **Modus Ponens:** $P, P \rightarrow Q \vdash Q$
2. **Modus Tollens:** $\neg Q, P \rightarrow Q \vdash \neg P$
3. **Hypothetical Syllogism:** $P \rightarrow Q, Q \rightarrow R \vdash P \rightarrow R$
4. **Resolution:** $P \vee Q, \neg P \vee R \vdash Q \vee R$

Bayes' Theorem

$$P(H|E) = P(E|H) \times P(H) / P(E)$$

- $P(H)$ = Prior probability
- $P(E|H)$ = Likelihood
- $P(H|E)$ = Posterior probability
- $P(E)$ = Evidence (normalizing constant)

✓ Chapter 4 Checklist

- ☐ Build truth tables for all 5 connectives

- ☐ Identify tautologies, contradictions, contingencies
 - ☐ Translate 5 English sentences to FOPL
 - ☐ Use $\forall$ and $\exists$ correctly, explain their difference
 - ☐ Apply De Morgan's laws (both propositional and quantifier versions)
 - ☐ Apply Modus Ponens and Modus Tollens
 - ☐ Explain what unification is with an example
 - ☐ Trace the RRS (Resolution Refutation) procedure step by step
 - ☐ Apply Bayes' theorem to a medical diagnosis problem
 - ☐ Explain what a Bayesian Network is and draw a simple one
 - ☐ Explain forward vs. backward chaining with an example
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Navigation

← **Previous:** [Chapter 3 — Search Techniques](#)

→ **Next:** [Chapter 5 — Structured Knowledge](#)

 **Home:** [README](#)